



JKBOSE 2026

CLASS 11TH

BINOMIAL THEOREM

MATHS

LECTURE-02

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Exercise Discussion 7.1

2

Related Problems

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n b^n$$

$$\Rightarrow (a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r \rightarrow \text{Binomial Theorem.}$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r.$$

r^{th} term from the end is the $(n-r+2)^{\text{th}}$ term from the beginning.

EXAMPLE

Determine whether the expansion of $\left(x^2 - \frac{2}{x}\right)^{18}$ will contain a term containing x^{10} ?

(Sol) Here General Term;

$$T_{r+1} = {}^{18}C_r (x^2)^{18-r} \left(-\frac{2}{x}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{18}C_r x^{36-2r} \frac{(-2)^r}{x^r}$$

$$\Rightarrow T_{r+1} = {}^{18}C_r (-2)^r x^{36-3r}$$

For a Term Containing x^{10} , we have

$$36 - 3r = 10$$

$$\Rightarrow 3r = 36 - 10$$

$$\Rightarrow 3r = 26$$

$$\Rightarrow r = \frac{26}{3} \notin \mathbb{N}$$

This shows that the expansion of $\left(x^2 - \frac{2}{x}\right)^{18}$ will not contain a term containing x^{10} . Hence the result.

QUESTION (NCERT EXAMPLAR)

The total number of terms in the expansion of $(x + a)^{51} - (x - a)^{51}$ after simplification is

- 1** 102 ✗
- 2** 25 ✗
- 3** 26 ✓
- 4** None of these ✗

QUESTION (NCERT EXAMPLAR)

If the coefficients of x^7 and x^8 in $\left(2 + \frac{x}{3}\right)^n$ are equal, then n is

1 ☒ 56

2 ☒ 55

3 ☒ 45

4 ☒ 15

(Sol) Here $T_{r+1} = {}^nC_r (2)^{n-r} \left(\frac{x}{3}\right)^r$

$$\Rightarrow T_{r+1} = {}^nC_r \frac{2^{n-r}}{3^r} \cdot x^r$$

So, Co-efficient of $x^r = {}^nC_r \frac{2^{n-r}}{3^r}$

Here, Co-efficient of $x^7 = {}^nC_7 \frac{2^{n-7}}{3^7}$

Co-efficient of $x^8 = {}^nC_8 \frac{2^{n-8}}{3^8}$

Therefore ${}^nC_7 \frac{2^{n-7}}{3^7} = {}^nC_8 \frac{2^{n-8}}{3^8}$

$$\Rightarrow \frac{n!}{(n-7)!7!} \frac{2^{n-8} \times 2}{3^7} = \frac{n!}{(n-8)!8!} \frac{2^{n-8}}{3^7 \times 3}$$

$$\Rightarrow \frac{2}{(n-7)(n-8)!7!} = \frac{1}{(n-8)!8 \times 7! \times 3}$$

$$\Rightarrow 48 = n-7$$

$$\Rightarrow n = 55$$

Hence the result.

EXAMPLE

$$\frac{\sqrt{x}}{\sqrt{3}} = \sqrt{\frac{x}{3}} = \left(\frac{x}{3}\right)^{\frac{1}{2}}$$



Find the term independent of x in the expansion of $\left(\frac{\sqrt{x}}{\sqrt{3}} + \frac{\sqrt{3}}{2x^2}\right)^{10}$.

(Sol) Here General Term is:

$$T_{r+1} = {}^{10}C_r \left(\frac{x}{3}\right)^{\frac{1}{2}(10-r)} \left(\frac{\sqrt{3}}{2x^2}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{10}C_r \frac{x^{\frac{10-r}{2}}}{3^{\frac{10-r}{2}}} \times \frac{3^{\frac{r}{2}}}{2^r x^{2r}}$$

$$\Rightarrow T_{r+1} = {}^{10}C_r \frac{3^{\frac{r}{2}}}{3^{\frac{10-r}{2}}} \times \frac{1}{2^r} \cdot x^{\frac{10-r}{2}-2r}$$

$$\Rightarrow T_{r+1} = {}^{10}C_r \frac{3^{\frac{r}{2} - \frac{(10-r)}{2}}}{2^r} \cdot x^{\frac{10-5r}{2}}$$

$$\Rightarrow T_{r+1} = {}^{10}C_r \frac{3^{r-5}}{2^r} \cdot x^{\frac{10-5r}{2}}$$

For term independent of x , we have

$$\frac{10-5r}{2} = 0$$

$$\Rightarrow r = 2$$

Therefore, the Required Term

$$= {}^{10}C_2 \frac{3^{2-5}}{2^2}$$

$$\begin{aligned} &= \frac{10!}{8!2!} \times \frac{1}{4} \times \frac{1}{27} \\ &= \frac{5 \times 4 \times 3 \times 2 \times 1}{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} \times \frac{1}{4} \times \frac{1}{27} \\ &= \frac{5}{12} \end{aligned}$$

Hence the result.

QUESTION (JEE-MAINS 2023)

The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is

1

8 ~~X~~

$$(Sol) T_{r+1} = {}^5C_r (2x^3)^{5-r} \left(-\frac{1}{3x^2}\right)^r$$

2

9 ~~X~~

$$\Rightarrow T_{r+1} = {}^5C_r 2^{5-r} (x^3)^{5-r} \left(-\frac{1}{3}\right)^r \times \frac{1}{x^{2r}}$$

3

$\frac{80}{9}$ ✓

$$\Rightarrow T_{r+1} = {}^5C_r 2^{5-r} \left(-\frac{1}{3}\right)^r x^{15-5r}$$

$$Put \ 15-5r = 5$$

$$\Rightarrow 5r = 10$$

$$\Rightarrow r = 2$$

4

$\frac{26}{3}$ ~~X~~

So Required Co-efficient

$$= {}^5C_2 2^{5-2} \left(-\frac{1}{3}\right)^2$$

$$= \frac{5!}{3!2!} \frac{8}{9}$$

$$= \frac{5 \times 4 \times 3!}{3!2} \times \frac{8}{9} = \frac{80}{9}$$

EXAMPLE

Which is larger $(1.01)^{1000000}$ or 10,000?

(Sol) we have

$$(1.01)^{1000000} = (1 + 0.01)^{1000000}$$

$$\Rightarrow (1.01)^{1000000} = {}^{1000000}C_0 (1)^{1000000} + {}^{1000000}C_1 (0.01) + \text{Some other positive Terms.}$$

$$\Rightarrow (1.01)^{1000000} = 1 + 1000000 \times \frac{1}{100} + \text{Some other positive Terms.}$$

$$\Rightarrow (1.01)^{1000000} = 1 + 10,000 + \text{Some other positive Terms.} > 10,000$$

$$\Rightarrow (1.01)^{1000000} > 10,000. \quad \text{Hence the result.}$$

EXAMPLE



Using binomial theorem, prove that $6^n - 5n$ always leaves remainder 1 when divided by 25

Proof: From Binomial Theorem;

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n b^n.$$

Put $a=1$, $b=5$, we get;

$$(1+5)^n = {}^nC_0 (1)^n + {}^nC_1 (1)^{n-1} (5) + {}^nC_2 (1)^{n-2} (5)^2 + \dots + {}^nC_n (5)^n.$$

$$\Rightarrow 6^n = 1 + 5n + {}^nC_2 (5)^2 + {}^nC_3 (5)^3 + {}^nC_4 (5)^4 + \dots + {}^nC_n (5)^n$$

$$\Rightarrow 6^n - 5n = 5^2 \underbrace{\left\{ {}^nC_2 + {}^nC_3 (5) + {}^nC_4 (5)^2 + \dots + {}^nC_n (5)^{n-2} \right\}}_q + 1$$

$$a = bq + r$$

$$\Rightarrow 6^n - 5n = 25q + 1.$$

This shows that $6^n - 5n$ leaves the remainder 1 when divided by 25.

Hence proved.

QUESTION (NCERT EXAMPLAR)

If $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \dots + a_{2n}$ equals.

1 $\frac{3^n + 1}{2}$ ✓

2 $\frac{3^n - 1}{2}$ ✗

3 $\frac{1 - 3^n}{2}$ ✗

4 $3^n + \frac{1}{2}$ ✗

(Sol) Given $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$

Put $x = -1$, we get

$$3^n = a_0 - a_1 + a_2 - a_3 + a_4 - \dots - a_{2n-1} + a_{2n} \quad \text{--- (1)}$$

Put $x = 1$, we get;

$$1 = a_0 + a_1 + a_2 + a_3 + a_4 + \dots + a_{2n-1} + a_{2n} \quad \text{--- (2)}$$

Adding (1) & (2), we get;

$$3^n + 1 = 2(a_0 + a_2 + a_4 + \dots + a_{2n})$$

$$\Rightarrow a_0 + a_2 + a_4 + \dots + a_{2n} = \frac{3^n + 1}{2}$$

QUESTION (NCERT EXAMPLAR)

The coefficient of x^p and x^q (p and q are positive integers) in the expansion of $(1+x)^{p+q}$, are

- 1 Equal ✓
- 2 ✗ Equal with opposite signs
- 3 ✗ Reciprocal of each other
- 4 ✗ None of these

(Sol) Here $T_{r+1} = {}^{p+q}C_r x^r$.

So, Co-efficient of $x^p = {}^{p+q}C_p = \frac{(p+q)!}{p!(p+q-p)!} = \frac{(p+q)!}{p!q!}$

Co-efficient of $x^q = {}^{p+q}C_q = \frac{(p+q)!}{(p+q-q)!q!} = \frac{(p+q)!}{p!q!}$

QUESTION (NCERT EXAMPLAR)

$$1+2+3+\dots+n = \frac{n(n+1)}{2}$$

The number of terms in the expansion of $(a+b+c)^n$, $n \in \mathbb{N}$ is,

1 ✓ $\frac{(n+1)(n+2)}{2}$

2 ✗ $n+1$

3 ✗ $n+2$

4 ✗ $(n+1)n$

(Sol) $(a+b+c)^n = (a+t)^n$, where $t=b+c$.

$$= {}^nC_0 a^n + {}^nC_1 a^{n-1} t + {}^nC_2 a^{n-2} t^2 + \dots + {}^nC_n t^n$$

$$= \underbrace{{}^nC_0 a^n}_1 + \underbrace{{}^nC_1 a^{n-1} (b+c)}_2 + \underbrace{{}^nC_2 a^{n-2} (b+c)^2}_3 + \dots + \underbrace{{}^nC_n (b+c)^n}_{(n+1)}$$

So, the total number of Terms = $1+2+3+\dots+n+n+1$
 $= \frac{(n+1)(n+2)}{2}$

Number of terms in Multinomial

$(x_1 + x_2 + x_3 + \dots + x_k)^n$ is $n+k-1$
 C_{k-1}

$(a+b+c)^n$

$n+3-1$

C_{3-1}

$$= {}^{n+2}C_2 = \frac{(n+2)(n+1)\cancel{n!}}{2!}$$

$$= \frac{(n+1)(n+2)}{2}$$

QUESTION (NCERT EXAMPLAR)

The ratio of the coefficient of x^{15} to the term independent of x in $\left(x^2 + \frac{2}{x}\right)^{15}$ is

1

12 : 32

HP/W

2

1 : 32

3

32 : 12

4

32 : 1

QUESTION (NCERT EXAMPLAR)

If the seventh terms from the beginning and the end in the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ are equal, then n equals_____.

Homework.

$$T_7 = T_{6+1}$$

$$T_7' = T_{n-7+2} = T_{n-5} = T_{(n-6)+1}$$

$$(a+b)^n = n_{c_0} a^n + n_{c_1} a^{n-1} b^1 + n_{c_2} a^{n-2} b^2 + \dots + n_{c_n} b^n.$$

$$= \underbrace{n_{c_0} a^{n-0} b^0}_{T_1} + \underbrace{n_{c_1} a^{n-1} b^1}_{T_2} + \underbrace{n_{c_2} a^{n-2} b^2}_{T_3} + \dots + \underbrace{n_{c_n} a^{n-n} b^n}_{T_{n+1}}$$

$$T_{r+1} = n_{c_r} a^{n-r} b^r //$$

4 → ②

Thank You

$$(x+a)^3 = x^3 + 3x^2a + 3xa^2 + a^3$$

$$(x-a)^3 = x^3 - 3x^2a + 3xa^2 - a^3$$

$$\begin{aligned} (x+a)^3 - (x-a)^3 &= \cancel{x^3} + 3x^2a + \cancel{3xa^2} + a^3 \\ &\quad - \cancel{x^3} + 3x^2a - \cancel{3xa^2} + a^3 \\ &= 6x^2a + 2a^3. \end{aligned}$$